

ADVIK ESWARAN

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EDUCATION

Princeton University

Princeton, NJ

Bachelor of Arts (A.B.), Geosciences (*Specialization in Geophysics*)

Expected June 2027

Minors: Applied and Computational Mathematics, Computer Science

RESEARCH EXPERIENCE

University of Washington, Department of Earth & Space Sciences

Seattle, WA (Remote)

Student Researcher

June 2022 – Present

Advisors: Dr. T.J. Fudge, Dr. Olivia Truax

- Developed a novel Ensemble Kalman Filter methodology for Antarctic climate reconstruction, incorporating both ice core records and climate model simulations
- Reconstructed 19th-20th century Antarctic snow accumulation and sea level contribution
- Identified optimal Antarctic shallow ice core drilling locations using an Ensemble Kalman Filter-based optimal sensor technique; the first time this has ever been used for Antarctic ice core site selection
- Presented case studies for drilling new ice cores at Crary Ice Rise and Lyddan Ice Rise, and redrilling the ITASE cores in West Antarctica

Princeton University / Center for Oldest Ice Exploration

Princeton, NJ

Student Researcher

January 2026 – Present

Advisor: Dr. John Higgins

- Running Community Ice Sheet Model simulations on multi-millennial year timescales, in order to identify optimal ice core drilling locations for obtaining the oldest ice
- Developing a method to reconstruct time-varying climate forcing inputs to the model, using existing ice core records

National Centre for Atmospheric Science

Leeds, United Kingdom

Student Researcher

June 2025 – August 2025

Advisor: Dr. Ryan Neely III

- Produced 2D spatial reconstructions of Antarctic snow accumulation using atmospheric radar profiles
- Developed a process-based snowfall retrieval scheme from a scanning Doppler radar at Troll Station, Antarctica

Juneau Icefield Research Program

Juneau, AK

Student Researcher

June 2024 – August 2024

Advisors: Dr. Eric Steig, Dr. Bradley Markle

- Contributed to numerous field glaciology research projects, including the drilling and processing of three ~15 m ice cores for water isotope sampling and installation of a TRI radar station to study supraglacial hydrology

PUBLICATIONS

Eswaran, A., Truax, O.J., & Fudge, T.J. (2024). 20th-Century Antarctic Sea Level Mitigation Driven by Uncertain East Antarctic Accumulation History. *Geophysical Research Letters*, 51, e2023GL106991. <https://doi.org/10.1029/2023GL106991>

Under Review

Eswaran, A., Truax, O.J., MacDonell, S., Matsuoka, K., Neff, P.D., Steig, E.J., Winton, V.H.L., & Fudge, T.J. (2026). Data-Driven Optimization of Antarctic Shallow Ice Core Site Selection. *Journal of Glaciology*. (Under Review)

CONFERENCE PRESENTATIONS

Eswaran, A., Truax, O., & Fudge, T.J. (2025), Optimal Sensor Analysis for Antarctic Ice Core Site Identification. 4th US Ice Core Community Meeting, Minneapolis, MN, 12-14 May. (Oral)

Fudge, T.J., Eswaran, A., & Truax, O. (2024), Optimal Sensor Analysis for Identification of Antarctic Ice Core Sites, C11C-0492 presented at 2024 Fall Meeting, AGU, Washington, DC, 9-13 Dec. (Poster)

Fudge, T.J., Eswaran, A., & Truax, O. (2024), Antarctic 20th-Century Sea Level Mitigation Dominated By Uncertainty in 19th-Century East Antarctic Snow Accumulation, EGU24-14163 presented at EGU General Assembly 2024, Vienna, Austria, 14-19 Apr. (Poster)

Eswaran, A., Truax, O., & Fudge, T.J. (2023), Late 20th-century Increase in Antarctic Snow Accumulation Drives Modest Mitigation of Sea Level Rise, C33E-1455 presented at 2023 Fall Meeting, AGU, San Francisco, CA, 11-15 Dec. (Poster)

Eswaran, A., Truax, O., & Fudge, T.J. (2023). Late 20th-century Increase in Antarctic Snow Accumulation Drives Modest Mitigation of Sea Level Rise. Northwest Glaciologists Meeting, Seattle, WA, 20-21 October. (Oral)

Eswaran, A. and Fudge, T.J. (2023), Reconstructing Antarctic Accumulation With Data Assimilation. 2nd US Ice Core Community Meeting, Seattle, WA, 8-10 May. (Poster)

Eswaran, A. and Fudge, T.J. (2022), Antarctic Ice Core Site Identification Using Data Assimilation of Synthetic Proxy Records for Twentieth Century Climate Reconstruction, C32D-0873 presented at 2022 Fall Meeting, AGU, Chicago, IL, 12-16 Dec. (Poster)

INVITED TALKS

“Optimal Sensor Analysis for Antarctic Shallow Ice Core Site Identification”, British Antarctic Survey, Cambridge, United Kingdom, 3 July 2025.

“Antarctic 20th-Century Sea Level Mitigation Dominated By Uncertainty in 19th-Century East Antarctic Snow Accumulation”, International Glaciological Society Global Seminar, 9 October 2024.

GRANTS/AWARDS

Student-Initiated Internship Award, High Meadows Environmental Institute, Princeton University, 2026

- Summer research visit funding to the University of Canterbury, Christchurch, New Zealand

Research Accelerator Grant, University of Washington Program on Climate Change, 2025

- Project funding for “Shallow Ice Core Drilling at Crary Ice Rise: A Pilot Project Proposal”

Student-Initiated Internship Award, High Meadows Environmental Institute, Princeton University, 2025

- Summer research visit funding to the National Centre for Atmospheric Science, Leeds, United Kingdom

Research Accelerator Grant, University of Washington Program on Climate Change, 2024

- Project funding for “Optimal Sensor Analysis for Identifying Shallow Antarctic Ice Core Drilling Locations”

Summer Undergraduate Research Award, NASA Space Grant, 2023

- Summer research award from the Washington NASA Space Grant Consortium

Outstanding Student Presentation Award, AGU Fall Meeting, 2022

- Chosen as one of the top six student presenters in the Cryosphere section

Summer Undergraduate Research Award, NASA Space Grant, 2022

- Summer research award from the Washington NASA Space Grant Consortium

SERVICE

Reviewer for *Geophysical Research Letters*

Reviewer for *The Cryosphere*

Co-Reviewer for *Nature Communications*

